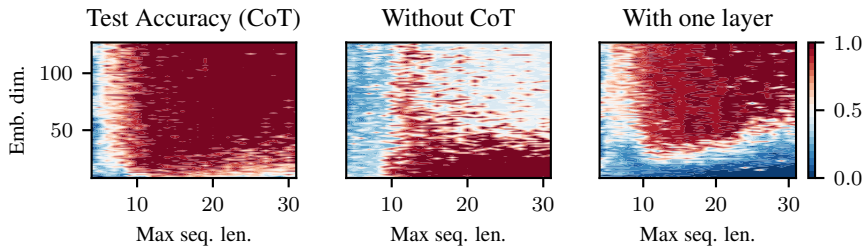

Figure 10: Same as Figure 7, except that we considered the parity problem, and 5000 training epochs.

iteration head circuit to solve the parity problem, allowing the parity problem to be learned within a very small number of epochs.

To deepen our understanding of the parity problem, we conducted the same scaling study depicted in Figure 7 using the parity dataset. The results are presented in Figure 10. Training was conducted over 5000 epochs. The data generation process was slightly modified to generate all sequences of length less than $L = \log_2(1048)$, and these were evenly split between training and testing (instead of generating redundant random sequences). In some sense, the parity problem can be considered relatively easy to solve in a non-iterative fashion: simply add all the elements of sequences, and reduce the sum modulo two. In theory, a single-layer transformer can use uniform attention to bring all the input tokens into superposition as input for the two-layer MLP layer, which is a universal approximator [27]. As a result, the parity with a fixed sequence length can be solved with such an architecture [see e.g. 5]. Indeed, the bottom right of the left and middle plots in Figure 10 indicate that next-token prediction performs better than chain-of-thought for sequence lengths up to $L_{\max} = 32$ with an embedding dimension of $d = 32$. This is due to the difficulty of performing position subtraction necessary for the CoT circuit, compared to the relative ease of performing addition of up to 32 bits with our two-layer architecture. Similarly, we found that a one-layer transformer was able to learn to produce correct CoT sequences for this task, demonstrating the existence of circuits fundamentally different from our iteration head to solve it. Anecdotally, the top right of the left and middle plots of Figure 10 indicate that as the model capacity increases, next-token prediction tends to overfit the training data, while CoT induces the transformer toward understanding the underlying structure that generated the data.

5 Conclusion

In this paper, we have explored the emergence of Chain-of-Thought (CoT) reasoning in Large Language Models (LLMs) through the lens of iterative algorithms. We have shown that, despite being trained on next-token prediction tasks, transformers can learn to solve iterative tasks efficiently using CoT reasoning. In particular, we have demonstrated that a two-layer transformer can implement what we named an "iteration head", enabling it to learn any iterative algorithm, assuming that it has enough feedforward layers following its two transformer blocks.

We have also shown that data curation can play a significant role in guiding the model towards the implementation of specific circuits. While our study has focused on simple, controlled problems and architectures, we hope that our findings shed light on the emergence of CoT capabilities in larger LLMs, whose attention patterns are much harder to interpret. In particular, they suggest that transformers are likely to develop "inner circuits" dedicated to multistep reasoning, which can then be applied to a variety of tasks that share the same underlying logical structure.

Interestingly, our work also highlights a limitation of the transformer architecture: they are stateless models. Indeed, our CoT implementation of Algorithm 1 requires the generated states (s_t) to have a token representation. This allows us to recover the state of the iterative algorithm at the root (i.e., the input) of the transformer. For complex iterative algorithms, or generic language modeling, it would be more logical to maintain a state internal to the model in the embedding space. The fact that GPT architectures do not allow this is arguably a shortcoming of the current transformer architecture [see 32, 6, 22, 47, 62, for interesting discussions], [see also 42, 33].

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